

E.3 Perceived Impacts

This appendix provides further detail on the perceived impacts reported in Section E.3. As a complement to the difference-in-differences analysis of labor market outcomes, respondents were asked directly: “*Have AI chatbots affected your labor earnings?*” If so, they were then asked: “*By how much?*”

While individuals may struggle to assess such counterfactuals precisely, we treat their responses as supplementary evidence based on a fundamentally different methodology that helps address some of the limitations of our difference-in-differences analysis.

First, as shown by Arcidiacono et al. (2020); Fuster, Kaplan and Zafar (2021); Giustinelli and Shapiro (2024), subjective assessments can reveal individual-specific treatment effects that may vary along unobservable dimensions not captured in survey or register data. In particular, our heterogeneity analyses above may fail to isolate the subset of workers most affected by AI chatbots.

Second, while difference-in-differences estimates identify only *differential* impacts among adopters, subjective assessments are less susceptible to this “missing intercept” problem. For example, if chatbot availability constitutes an economy-wide shock, it could affect earnings across entire occupations, regardless of individual or workplace adoption. While our estimates in Section 3.1 identify effects only up to the wage impact on non-encouraged non-adopters, we may still ask these workers about their perceived labor market impacts from AI chatbots. If substantial occupation-wide general equilibrium effects are unfolding, they may perceive the impact on their earnings even without adopting the tools themselves.

Finally, eliciting perceived earnings effects offers a bridge between the self-reported work benefits in Section D and the observed earnings outcomes in Section 3. This helps assess whether the weak link between perceived work-related benefits and observed earnings outcomes reflects misperception on the part of workers—or instead an accurate understanding that the pass-through from work-related gains to earnings is limited.

Table E.11 reports workers’ perceived earnings impacts from AI chatbots, with Figure

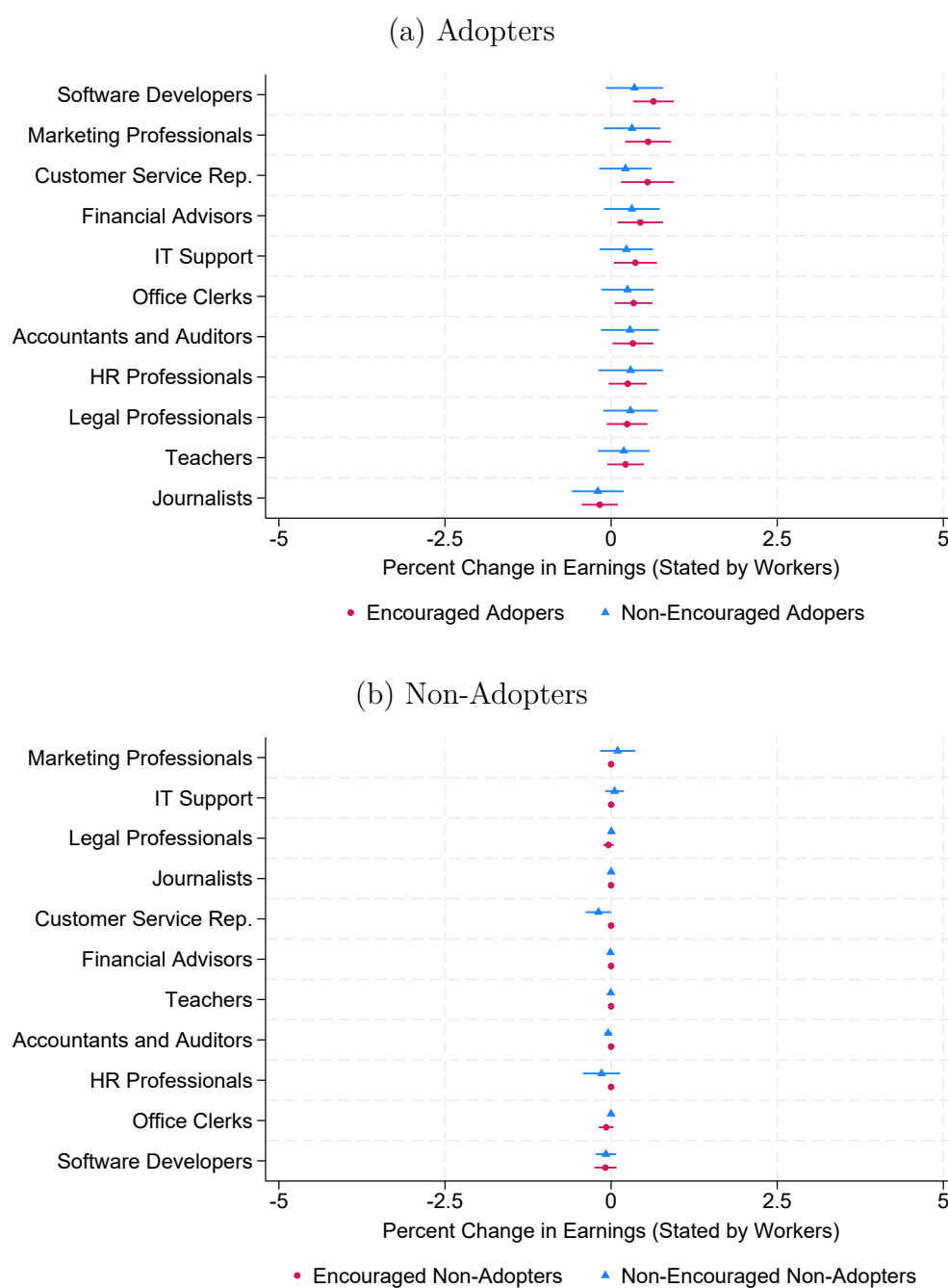
E.18 showing averages by occupation, employer encouragement, and adoption status. Nearly all respondents—97.7% of adopters and 99.5% of non-adopters—report *no* impact on their earnings. Consistent with this, Figure E.19 shows that workers themselves estimate only 3–7% of their time savings translate into higher earnings.

Table E.11: Decomposition of Perceived Earnings Effects of AI Chatbots Among Adopters

Occupation	Policy	Average Impact (in %) (1)	Share of Impacts			Conditional Average Impacts (in %)		
			Decreased Earnings (2)	Unchanged Earnings (3)	Increased Earnings (4)	Decreased Earnings (5)	Unchanged Earnings (6)	Increased Earnings (7)
Journalists	Non-Encouraged	-.118		.994			0	
Journalists	Encouraged	-.159		.983			0	
Software Developers	Non-Encouraged	.092	.008	.974	.015	-12.187	0	12.5
Software Developers	Encouraged	.428		.96	.037		0	11.944
Legal Professionals	Non-Encouraged	.031		.978			0	
Legal Professionals	Encouraged	.057		.987	.009		0	10.5
Accountants and Auditors	Non-Encouraged	0	.004	.915	.004	-13.75	0	13.75
Accountants and Auditors	Encouraged	.126		.973	.015		0	10
Customer Service Rep.	Non-Encouraged	-.11	.013	.913		-9.167	0	
Customer Service Rep.	Encouraged	.303		.942	.041		0	7.25
Marketing Professionals	Non-Encouraged	.112		.956	.023		0	7.25
Marketing Professionals	Encouraged	.37	.006	.955	.038	-16.5	0	12.422
Financial Advisors	Non-Encouraged	.022		.964			0	
Financial Advisors	Encouraged	.214		.979	.019		0	11.667
HR Professionals	Non-Encouraged	0		.964			0	
HR Professionals	Encouraged	.071		.986			0	
Office Clerks	Non-Encouraged	.015	.002	.901	.003	-7.143	0	8.462
Office Clerks	Encouraged	.113		.978	.014		0	10.403
Teachers	Non-Encouraged	-.007	.001	.975	.001	-9.167	0	4
Teachers	Encouraged	.041		.992	.007		0	7.188
IT Support	Non-Encouraged	.039	.006	.981		-4	0	
IT Support	Encouraged	.172		.974	.02		0	11.25

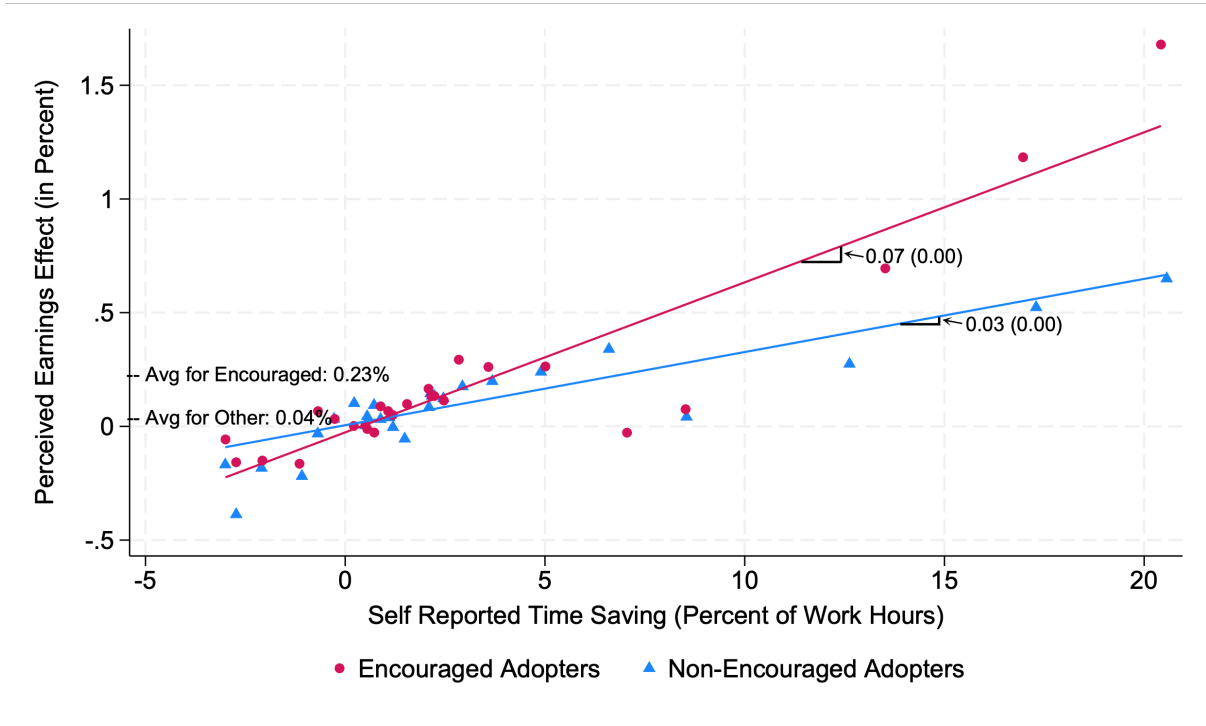
Notes: This table decomposes the average perceived earnings impact (in percent) of AI chatbots among adopters (Column 1) into contributions from workers reporting decreased earnings (Columns 2 and 5), unchanged earnings (Columns 3 and 6), and increased earnings (Columns 4 and 7). Responses are categorized by workers' occupations and whether their employers encourage AI chatbot use. Empty cells represent fewer than five individuals and are blanked out in accordance with confidentiality rules. We code reported earnings impacts as follows: “No change” as 0%, “0–5% change” as 2.5%, “5–15% change” as 10%, and “Over 15% change” as 20%. Our conclusions are robust to alternative codings. *Sample:* All completed responses from the 2024 survey round linked to registry data.

Figure E.18: Perceived Labor Market Impacts among Workers



Notes: This figure shows the average perceived impact of AI chatbots on earnings, broken down by workers' occupations and whether their employers encourage chatbot use. Panel (a) focuses on adopters—workers who have used AI chatbots for work. Panel (b) focuses on non-adopters—those who have not. See the notes of Table E.11 for how we code perceived earnings impacts. Whiskers represent 95% confidence intervals. *Sample:* All completed responses from the 2024 survey round linked to registry data.

Figure E.19: How Do Workers' Earnings Impacts Relate to Their Time Savings?



Notes: This figure presents a binned scatterplot of workers' perceived earnings impacts from AI chatbots against their estimated time savings from these tools, absorbing occupation fixed effects. The sample is divided based on whether employers encourage AI chatbot use. The regression line represents the line of best fit for each group, controlling for occupation fixed effects. *Sample:* All completed responses from the 2024 survey.